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Paper Code : OECCS701B Multimedia Systems

UPID : 007563

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer *any ten* of the following :

[1 x 10 = 10]

- (I) What is raster graphics?
- (II) MP3 produces three data rates from 96 Kbps to _____ Kbps.
- (III) List two type of multimedia.
- (IV) In Real-Time Transport Protocol (RTP), source periodically sends a source description message to give additional information about _____.
- (V) What is generative graphics ?
- (VI) If an image has a width of 4 inch and an aspect ratio of 3 : 2, then what is its height?
- (VII) Live streaming is still using Transmission Control Protocol (TCP), and multiple unicasting instead of _____.
- (VIII) State two disadvantages of microwave.
- (IX) In Real Time Interactive Audio Video, conferencing requires two way communication between _____
- (X) How many step processes are there to create a 3-D animation?
- (XI) In Real Time Interactive Audio Video, Jitter is introduced in real-time data by delay between _____
- (XII) Joint Photographic Experts Group (JPEG) is used to compress _____

Group-B (Short Answer Type Question)

Answer *any three* of the following :

[5 x 3 = 15]

2. How can a text be inserted within an application? Explain briefly. [5]
3. What is meant by continuous tone, half-tone and bitone images? Give Example. [5]
4. Define Sampling. What is meant by sampling rate? [5]
5. Write a short note on QBIC. [5]
6. Differentiate between formatted text and unformatted text. [5]

Group-C (Long Answer Type Question)

Answer *any three* of the following :

[15 x 3 = 45]

7. (a) How does the RGB color model represent color information? Why is it called an additive model? [5]
 (b) How does the CMYK color model represent color information? Why is it called a subtractive model? [5]
 (c) Discuss the components of multimedia. [5]
8. (a) How can you represent a digital image? [3]
 (b) Difference between captured image and stored image format. [5]
 (c) Explain different types of graphics format. [3]
 (d) Difference between sound and speech. [4]
9. (a) Discuss the characteristics of multimedia. [5]
 (b) Discuss the application of multimedia. [5]
 (c) Discuss the impact of multimedia in recent days. [5]
10. (a) Write a short note on kd trees. [5]
 (b) Write a short note on R-trees. [5]
 (c) Write a short note on Quad-trees. [5]
11. (a) Write a short note on CD-ROM. [5]
 (b) Define spatial and temporal redundancy. [5]

(c) Write a short note on DVD.

[5]

*** END OF PAPER ***